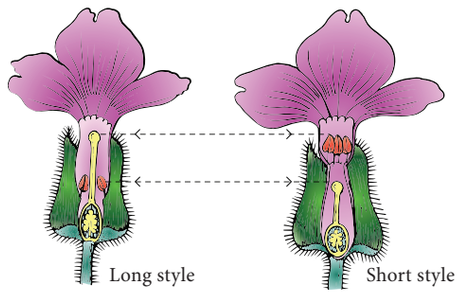
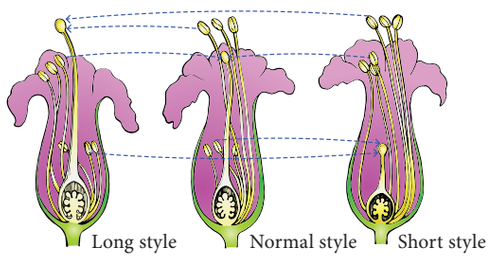
iii. Heterostyly: Some plants produce two or three different forms of flowers that are different in their length of stamens and style. Pollination will take place only between organs of the same length.(Figure 1.14)

a. Distyly: The plant produces two forms of flowers, Pin or long style, long stigmatic papillae, short stamens and small pollen grains; Thrum-eyed or short style, small stigmatic papillae, long stamens and large pollen grains. Example: *Primula* (Figure 1.14a). The stigma of the Thrum-eyed flowers and the anther of the pin lie in same level to bring out pollination. Similarly the anther of Thrum-eyed and stigma of pin ones is found in same height. This helps in effective pollination.



a) Distyly - *Primula*



b) Tristyly - *Lythrum*

Figure 1.14 Heterostyly

b. Tristyly: The plant produces three kinds of flowers, with respect to the length of the style and stamens. Here,the pollen from flowers of one type can pollinate only the other two types but not their own type. Example : *Lythrum* (Figure 1.14b).

iv. Self sterility/ Self- incompatibility: In some plants, when the pollen grain of a flower reaches the stigma of the same, it is unable to germinate or prevented to germinate on its own stigma. Examples: *Abutilon*, *Passiflora*. It is a genetic mechanism.

Agents of pollination

Pollination is effected by many agents like wind, water, insects etc. On the basis of the agents that bring about pollination, the mode of pollination is divided into abiotic and biotic. The latter type is used by majority of plants.

Abiotic agents

1. Anemophily - pollination by Wind
2. Hydrophily - pollination by Water

Biotic agent

1. Zoophily

Zoophily refers to pollination through animals.

Abiotic agents

1. Anemophily: Pollination by wind. The wind pollinated flowers are called **anemophilous**. The wind pollinated plants are generally situated in wind exposed regions. Anemophily is a chance event. Therefore, the pollen may not reach the target flower effectively and are wasted during the transit from one flower to another. The common examples of wind pollinated flowers are - grasses, sugarcane, bamboo, coconut, palm, maize etc.,

Anemophilous plants have the following characteristic features:

- The flowers are produced in pendulous, catkin-like or spike inflorescence.
- The axis of inflorescence elongates so that the flowers are brought well above the leaves.
- The perianth is absent or highly reduced.
- The flowers are small, inconspicuous, colourless, not scented, do not secrete nectar.
- The stamens are numerous, filaments are long, exerted and versatile.
- Anthers produce enormous quantities of pollen grains compared to number of ovules available for pollination. They are minute, light and dry so that they can be carried to long distances by wind.

TN_GOVT_BOTANY_XII_PAGES_001-032 CH 01.indd 15 23-12-2022 10:56:44